Survival-supervised deconvolution identifies a reproducible tumor–stroma prognostic axis in pancreatic cancer

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Abstract

Pancreatic ductal adenocarcinoma (PDAC) transcriptomes contain heterogeneous tumor and microenvironmental programs with differing relevance to survival, yet standard deconvolution methods learn these programs without using survival outcomes. We developed DeSurv, a survival-supervised matrix factorization framework in which survival helps shape gene programs as they are learned. DeSurv recovered a compact three-program tumor–stroma axis from pooled TCGA and CPTAC cohorts, with classical/restCAF-like programs at one pole and basal-like/proCAF-like programs at the higher-risk pole. The axis reflects coordinated tumor–stroma variation across patients, rather than spatial or cellular organization within individual tumors. Across five independent cohorts, movement toward the basal-like/proCAF pole was associated with worse survival (HR per SD, 1.50; 95% CI, 1.31–1.72) and remained prognostic after PurIST and DeCAF adjustment. Despite predictive performance comparable to supervised models, DeSurv resolved the prognostic signal into a compact, reproducible tumor–stroma representation, including a distinct stromal program not separately resolved by those models.

Pancreatic ductal adenocarcinoma (PDAC) is one of the most lethal solid tumors, with few durable responses to existing therapies[1]. Bulk transcriptomic studies have identified clinically relevant malignant and stromal states in PDAC, including basal-like and classical tumor programs[2–4] and tumor-promoting proCAF and tumor-restraining restCAF programs[5]. Single-cell and spatial profiling have further shown that this heterogeneity extends across malignant, stromal, immune, and other microenvironmental compartments within individual tumors[6–8]. Whether these tumor and stromal programs together provide prognostic information beyond existing subtype classifications remains unclear.

Although single-cell and spatial profiling resolve cellular states in detail, large clinically annotated PDAC cohorts still rely primarily on bulk transcriptomic profiling[9,10]. Bulk profiles provide the scale needed for survival analysis but mix signals from malignant and microenvironmental cells[11];

nonnegative matrix factorization (NMF) and related deconvolution methods can separate these composite profiles into interpretable gene programs[3,12–14]. In PDAC, such methods have yielded molecular subtypes[2,4,15], virtual microdissection of basal-like and classical malignant programs[3], and compartment-specific tumor–stroma deconvolution[5,14].

These deconvolution approaches typically discover gene programs without using clinical outcomes and assess their prognostic relevance only afterward[2,15]. Standard NMF learns gene programs by minimizing reconstruction error, favoring patterns that capture prominent expression variation whether or not that variation is related to survival[9,16]. This distinction is particularly important in PDAC, where bulk tumors contain substantial stromal, exocrine, and other non-malignant signals that can dominate transcriptomic variation and obscure less prominent but prognostically relevant programs[4,5,17,18]. Reconstruction alone also does not uniquely determine the recovered gene programs without additional constraints[19–21], which may contribute to cross-study variability among unsupervised PDAC subtype taxonomies[22].

Conversely, survival-supervised models can identify outcome-associated expression patterns, but they often operate on thousands of highly correlated genes, making individual-gene effects difficult to estimate and interpret. Such models may predict survival without organizing the prognostic signal into biologically interpretable tumor and microenvironmental programs. Together, these limitations motivate an approach that incorporates survival information as gene programs are learned, while preserving their biological interpretability.

Here, we present DeSurv, a survival-supervised deconvolution framework that integrates NMF with Cox regression so that survival helps shape the gene programs as they are learned. By incorporating survival during factorization, DeSurv is designed to recover compact, survival-aligned representations that may be obscured when other sources of expression variation dominate. Once learned, the programs are held fixed, allowing independent cohorts to test whether the same biological programs retain their association with survival. We evaluate DeSurv first in simulations, where the underlying prognostic programs are known, and then across five independent PDAC cohorts.

In PDAC, DeSurv identifies a compact three-program structure organized around a co-varying tumor–stroma prognostic axis, reflecting coordinated variation in tumor and stromal programs across patients rather than their spatial or cellular organization within individual tumors. Classical tumor identity and restCAF-like stroma characterize one pole, whereas basal-like tumor identity and proCAF-like stroma characterize the higher-risk pole. This organization recovers and extends prior evidence that tumor-intrinsic and stromal states jointly shape PDAC outcome, resolving their relationship as a continuous survival-associated axis. Across five independent PDAC cohorts, the axis remains associated with survival and provides prognostic information beyond established PurIST and DeCAF classifications. Although higher-rank NMF models achieve comparable survival prediction and external generalization, they require more than twice as many factors and distribute the relevant biology across a more complex representation rather than recovering the same compact tumor–stroma organization. In simulations, DeSurv more reliably recovers prognostic programs when they account for less expression variation than outcome-neutral programs. Together, these findings show that survival supervision can recover a compact, reproducible, and biologically interpretable representation of prognostically relevant tumor–stroma variation.

Results

DeSurv learns survival-associated gene programs from bulk transcriptomes

DeSurv learns gene programs that balance reconstruction of the tumor transcriptome with association to survival. After training, the learned gene-program basis (W) is fixed, allowing the same programs to be scored directly in new tumors without refitting the factorization (Fig. 1A; Methods). We trained DeSurv in pooled TCGA and CPTAC PDAC cohorts[23,24] ($n = 273$, 139 events). Cross-validated concordance changed little beyond three programs, and Bayesian optimization with the one-standard-error rule selected a parsimonious three-program model ($k = 3$, $\alpha = 0.334$; peak cross-validated C-index, 0.655; Fig. 1B and Supplementary Fig. S4). By contrast, standard unsupervised NMF rank criteria (reconstruction residuals, cophenetic correlation and silhouette width[13,25]) gave conflicting choices in the same data (Supplementary Fig. S5), underscoring rank selection as an additional source of ambiguity in unsupervised decompositions. All model and hyperparameter selection was completed within the training cohorts before the programs were applied to external validation datasets.

Survival supervision resolves a distinct tumor–stroma program structure

To determine how survival supervision altered the biological representation, we compared DeSurv with standard NMF at the same rank ($k = 3$), thereby holding model complexity constant (a sensitivity analysis using NMF's independently selected ranks, elbow $k = 5$ and BO $k = 7$, is in Supplementary Table S5). DeSurv resolved three distinct programs (D1–D3): D1, enriched for classical tumor and restCAF-like stromal signatures; D2, enriched for proCAF-like stroma; and D3, enriched for basal-like tumor programs (Fig. 2A). Together, these programs define a tumor–stroma prognostic architecture spanning classical/restCAF-like and basal-like/proCAF-like states across patients. The classical identity of D1 was independently supported in the COMPASS cohort, where D1 scores correlated with GATA6 RNA in situ hybridization (Spearman $\rho = 0.68$, $P < 0.001$, $n = 33$; Fig. 2E).

Standard NMF produced a different organization at the same rank (Fig. 2B). Its malignant factor (N1) combined classical and basal-like tumor programs, its microenvironmental factor (N3) combined multiple stromal and immune programs[26,27], and a third factor (N2) captured exocrine-associated variation. DeSurv instead separated basal-like tumor (D3) and proCAF-like stroma (D2) while recovering the distinct classical/restCAF-like survival-aligned D1 program.

This difference reflected a selective reorganization rather than wholesale loss of expression structure. D1 accounted for only 18.8% of reconstruction contribution (normalized Shapley value; Fig. 2C) but nearly all of the factor-specific survival contribution in the training data ($\Delta\ell = 66.4$, the change in Cox partial log-likelihood on factor removal), illustrating that the programs that best reconstruct the transcriptome need not be those most relevant to survival[18]. Pairwise gene-loading correlations across all 1{,}970 genes showed that the principal NMF tumor and microenvironmental factors were largely retained as D3 (Spearman $\rho = 0.84$) and D2 ($\rho = 0.87$), whereas D1 correlated only weakly with every matched-rank NMF factor ($\rho \leq 0.36$; Fig. 2D). Consistently, D1 was poorly reconstructable from the full NMF basis ($R^2 = 0.09$), while DeSurv sacrificed only about 1.5 percentage points of whole-transcriptome reconstruction R^2 (Supplementary Table S2). Thus, survival supervision largely preserved variance-dominant tumor and stromal structure while reorganizing the representation to expose a distinct survival-aligned program.

Survival-aligned programs generalize across independent PDAC cohorts

We next tested whether the programs learned in TCGA and CPTAC generalized without retraining to five independent PDAC cohorts (Dijk, Moffitt, PACA-AU array, PACA-AU RNA-seq and Puleo; non-metastatic, treatment-naive; Supplementary Table S1; $n = 616$, 414 events), projecting the trained programs into each validation expression matrix ($Z = \widetilde{W}^{\top} X_{\text{new}}$). The frozen DeSurv linear predictor (oriented so that higher values indicate higher risk) was associated with overall survival in a cohort-stratified analysis (pooled HR per SD = 1.50, 95% CI 1.31–1.72, $P = 2.6 \times 10^{-9}$; Fig. 3A). At the same three-program complexity, the corresponding standard NMF predictor showed a substantially weaker association (HR per SD = 1.11, 95% CI 1.00–1.23, $P = 0.057$).

We then asked whether comparable prognostic performance could be achieved without survival-supervised deconvolution. Increasing the rank of standard NMF substantially improved external discrimination: NMF at $k = 7$ matched or exceeded DeSurv's concordance in most validation cohorts and, like DeSurv, remained prognostic after simultaneous adjustment for PurIST and DeCAF (Table 1; Supplementary Table S5). Thus, survival supervision was not required to recover prognostic information from the transcriptome. Increasing NMF rank, however, distributed this information across a rank-sensitive set of overlapping factors: at $k = 7$, multiple factors shared the same established PDAC signatures and the additional factors contributed little individually to survival despite strong performance of the combined predictor (Supplementary Fig. S10).

As a secondary, clinical visualization, a risk cutpoint selected entirely within the training data separated survival in the pooled external validation cohort (HR = 1.79, 95% CI 1.41–2.28, $P < 0.001$; Fig. 3B); per-cohort Kaplan–Meier analyses are provided in the Supplementary Information. Prediction alone therefore did not distinguish DeSurv from higher-rank NMF, and, as we show next, comparable prediction is also achieved by conventional survival-supervised models. We therefore asked whether DeSurv instead provided a different biological representation of the prognostic signal, beyond that available from established classifiers and other survival-supervised approaches.

Table 1: Pooled external validation hazard ratios (per SD of the linear predictor) for DeSurv at $k = 3$ and standard NMF at BO-selected $k = 7$ ($\alpha = 0$), across the five validation cohorts ($n = 616$ patients, 414 events). Adjusted models include the PurIST tumor-intrinsic and DeCAF stromal molecular classifiers as covariates, with stratification by dataset. Both methods retain significant prognostic value after adjustment.

Method	Unadjusted HR per SD (95% CI), P	Adjusted HR per SD (95% CI), P
DeSurv ($k = 3$)	1.50 (1.31–1.72), < 0.001	1.33 (1.13–1.56), < 0.001
Std NMF ($k = 7$, BO $\alpha = 0$)	1.48 (1.34–1.64), < 0.001	1.47 (1.27–1.70), < 0.001

The survival-aligned structure is reproducible and complementary to established PDAC classifiers

We next asked whether the dominant D1 direction simply recapitulated established PDAC subtype classifications. Across the pooled validation cohorts ($n = 616$, 414 events), binary PurIST and DeCAF classifications together explained 28% of the within-cohort variance in the continuous D1 score, indicating substantial alignment with known tumor-intrinsic and stromal subtypes but incomplete overlap. D1 remained associated with survival after simultaneous adjustment for both classifiers (HR per SD = 0.82, 95% CI 0.74–0.92, $P = 0.0006$); adding D1 to a stratified Cox model containing PurIST and DeCAF improved model fit (partial likelihood-ratio $\chi^2 = 11.6$ on 1 df, $P = 0.0006$) and concordance from 0.593 to 0.621. The DeSurv linear predictor also remained prognostic after adjustment for standard clinical covariates and, where adequately powered, for tumor purity (Supplementary Table S4). Thus, the survival-aligned representation overlaps established tumor and stromal classifications without being reducible to them.

The dominant D1 direction was also not specific to the DeSurv optimizer. Three independent survival-supervised approaches—supervised principal components, Cox partial least squares and sparse Cox—recovered patient scores strongly correlated with D1 in the training cohort ($|r| = 0.81$, 0.83 and 0.85, respectively), whereas the leading unsupervised principal component was nearly orthogonal to D1 ($|r| = 0.10$) and instead tracked higher-variance D2/D3 structure (Supplementary Information, supervised-method recovery). Thus, D1 reflects a reproducible survival-associated expression direction shared across supervised approaches rather than an idiosyncrasy of DeSurv.

Having established D1 as a reproducible survival-associated direction, we asked what DeSurv adds beyond recovering it. Independently tuned supervised principal components and penalized Cox achieved pooled external concordance comparable to DeSurv (C-index 0.60, 0.61 and 0.61, respectively) and captured the dominant D1-associated prognostic direction, but were nearly orthogonal to the proCAF-associated D2 program ($|r| = 0.06$ and 0.14; Fig. 3C). Their selected genes consistently enriched basal-like tumor but not proCAF programs; although their risk scores retained modest stromal signal, they did not resolve it as a distinct stromal program (Supplementary Information). By jointly preserving transcriptomic reconstruction while allowing survival to reshape the learned basis, DeSurv instead organized the transcriptomic context around the dominant prognostic direction into distinct classical/restCAF-like, proCAF-like and basal-like programs. This organization is consistent with DeSurv retaining major expression structure while reorganizing it around prognosis, rather than optimizing prediction or reconstruction alone. Importantly, it converts a single prognostic direction into biologically attributable tumor and stromal programs that can be independently validated and interrogated.

DeSurv recovers prognostic gene programs when variance and prognosis diverge

We used simulations with known latent structure to define when survival supervision alters program recovery. In the primary scenario, the gene program driving survival contributed less transcriptomic variation than outcome-neutral programs ($p = 3{,}000$ genes, $n = 200$ samples, true $k = 3$; Materials and Methods, Simulation Studies), so that the strongest sources of expression variation were not the most prognostically relevant. DeSurv and standard NMF ($\alpha = 0$) were tuned using the same cross-validation procedure, isolating the effect of survival supervision during factorization.

Under this setting, DeSurv more accurately recovered the genes defining the true prognostic program, whereas standard NMF largely followed variance-dominant structure (Fig. 4A–B). This distinction is important because the genes assigned to a recovered program—not only its ability to rank patients by risk—determine its subsequent biological interpretation and its potential use in candidate signatures or biomarker development. Across 100 replicates, median test-set C-index was 0.837 for DeSurv versus 0.724 for standard NMF ($\Delta = 0.113$, paired Wilcoxon $P < 0.001$), and DeSurv recovered the true factorization rank ($k = 3$) more frequently (Fig. 4C).

The advantage depended on the relationship between transcriptomic variance and prognosis. It attenuated when prognostic and variance-dominant structure partially overlapped and disappeared under the null, where neither method identified prognostic signal (Supplementary Fig. S3). In the mixed setting, the advantage remained strongest for recovery of the factor-specific marker genes rather than shared background survival genes, supporting a specific effect of supervision on identifying the genes that define the prognostic program. Thus, survival supervision is most useful for biological discovery when clinically relevant programs are not themselves the dominant sources of transcriptomic variation.

Discussion

Survival-supervised factorization revealed that the dominant prognostic transcriptional structure in PDAC is a tumor–stroma axis, coupling classical tumor identity and restCAF-like stroma at one pole with basal-like tumor identity and proCAF-like stroma at the other. This axis generalized in a pooled analysis of five independent PDAC cohorts without retraining and remained prognostic after adjustment for PurIST and DeCAF, indicating that outcome is associated with the coordinated configuration of tumor-intrinsic lineage state and stromal composition, not fully captured by the compartment-specific classifiers evaluated here. Although purely supervised predictors fit directly to bulk expression achieved similar external discrimination, they returned a single, compartment-ambiguous risk score. This held even for canonical, independently tuned supervised methods, which recovered the tumor-axis direction but expressed it as an unstable, basal-like tumor gene set; although these scores retained modest stromal signal, they did not resolve it as a distinct stromal program, unlike DeSurv's separate stromal factor (Supplementary Information). DeSurv instead decomposed this prognostic signal into a small number of interpretable, source-attributable programs by coupling NMF to a Cox objective during program discovery.

This distinction is methodological as well as biological. In DeSurv, survival supervision acts on the transferable gene-program basis, W, while preserving the interpretation of sample loadings, H, as mixture coefficients, so the learned programs project into new samples without re-estimating the factorization. Prior survival-supervised NMF approaches have similarly integrated Cox objectives with matrix factorization, but generally supervise the latent sample representation or impose survival constraints on the factorization as a whole[28,29]. Deep survival-representation methods, including interpretable autoencoder- and attention-based Cox models, can learn prognostic latent features but typically return representations whose biological attribution depends on model-specific decoding or explanation steps[30,31]. DeSurv's contribution is to keep the prognostic objective tied to a fixed, source-attributable gene-program basis, making the recovered programs both interpretable and directly transportable across cohorts.

Biologically, DeSurv aligns the tumor–stroma axis with a single survival-associated program: the factor with the largest survival contribution, D1, links classical tumor and restCAF-like stromal genes, recovered de novo without predefined signatures while their survival associations are estimated during factorization. This organization is concordant with virtual microdissection[3], experimental microdissection[27], and unsupervised deconvolution[14], and with evidence that PDAC outcome is shaped by tumor-intrinsic state together with fibroblast and stromal programs[5,32–34] whose cellular states are themselves heterogeneous[35]. By contrast, standard NMF at the matched rank of $k = 3$ devoted one factor to exocrine-associated, low-purity variation[4], whereas DeSurv separated the microenvironmental programs at the selected rank.

This parsimony reflects both survival supervision and the model-selection framework. Survival supervision produced a broad concordance plateau across ranks, letting the one-standard-error rule select a compact three-factor model that standard NMF matched only at higher rank and greater complexity (Supplementary Table S5); standard NMF's own unsupervised rank-selection heuristics did not identify this solution (Supplementary Fig. S5). Joint optimization over rank and supervision strength, coupled to the one-standard-error rule, is therefore part of DeSurv's methodological contribution.

Our simulations clarify when this tradeoff is most useful. DeSurv's advantage was greatest when prognostic programs explained modest variance relative to outcome-neutral signals, and was absent under survival-null conditions. DeSurv is therefore not intended to replace unsupervised NMF in all settings. For exploratory analyses without clinical endpoints, settings with sparse or unreliable survival data, or studies focused on comprehensive biological characterization, unsupervised methods remain appropriate. The supervision weight, selected by Bayesian optimization, allows the data to determine the balance between reconstruction and survival association; at $\alpha = 0$, DeSurv reduces to standard NMF.

The same scope clarifies what DeSurv is not. It is not proposed as a competing molecular classifier: the relevant question is whether its factors add prognostic information beyond established classifiers, which they did after adjustment for PurIST and DeCAF (adjusted HR = 1.33, 95% CI 1.13–1.56, $P < 0.001$; Table 1). DeSurv is also distinct from reference-based deconvolution methods such as CIBERSORTx[36] and BayesPrism[37], which estimate cellular composition from single-cell or sorted-cell references; DeSurv instead learns prognostic latent gene programs directly from bulk expression without a cellular reference, and could be integrated with deconvolution-derived cell fractions to refine compartmental attribution.

Several limitations define the scope of these findings. DeSurv assumes proportional hazards, which may not hold under delayed or non-proportional effects, and the convergence guarantee applies to an idealized algorithm (Supplementary Theorem 1). Extensions to more flexible survival objectives are therefore a natural direction. The method was also applied here only to PDAC. Thus, although our simulations show that survival supervision can recover prognostic programs when variance-dominant and outcome-relevant subspaces diverge, the generality of this principle remains to be established across cancer types. Such studies will require multi-cancer benchmarking, including comparisons with strong penalized-regression baselines, which learned representations do not consistently outperform for out-of-sample prognosis[38].

The training and primary validation cohorts were non-metastatic and treatment-naive, so DeSurv was learned primarily in resectable disease. Projection of the trained programs into small treated cohorts ($n = 66$–85) suggested a consistent protective direction for the proCAF-like program and a more setting-dependent basal–classical axis, consistent with emerging evidence that PDAC stroma can have both tumor-promoting and tumor-restraining roles[39–41]. However, treatment, stage and cohort were confounded in these analyses, and the treated cohorts were small. We therefore regard these observations as exploratory and hypothesis-generating. Whether the proCAF-like program identifies patients who benefit from stroma- or myeloid-directed therapy will require prospective testing.

Finally, the tumor–stroma axis is defined from bulk gene programs: it reflects cross-patient covariation of classical-tumor and restCAF-like stromal signatures, not per-cell co-expression or spatial adjacency. Single-cell and spatial assays are better suited to resolve these relationships at cellular resolution. The bulk association is nevertheless supported by several orthogonal observations: D1 remained prognostic after PurIST and DeCAF adjustment, was recovered by independent survival-supervised methods but not by the leading unsupervised direction, and its classical-tumor identity was corroborated by GATA6 RNA in situ hybridization. Where tumor-purity annotations were available, purity adjustment further argued against the axis being driven solely by tumor purity. Adequately powered single-cell and spatial profiling of the same clinical settings is therefore an important next validation step, and mechanistic experiments will be needed to determine whether these tumor–stroma configurations causally shape PDAC progression.

More broadly, the conventional workflow of unsupervised discovery followed by retrospective clinical evaluation can lose prognostic structure when the dominant sources of expression variation are not those most relevant to outcome, as is common in bulk tumors where purity and tissue composition strongly shape transcriptomic profiles[18]; aligning factorization with the clinical question from the outset helps avoid this mismatch. Survival supervision also constrains the gene-program basis, which reconstruction alone leaves underdetermined and which may contribute to cross-study variability among unsupervised PDAC taxonomies[2,3,15]. Thus, survival supervision does more than improve prediction: it resolves the prognostic signal into reproducible, source-attributable gene programs, a property that distinguishes DeSurv from a conventional supervised risk score and should become increasingly useful as clinically annotated transcriptomic cohorts grow. Together, these findings support a model in which PDAC prognosis is associated with coordinated malignant–stromal transcriptional states that outcome-supervised deconvolution can recover from bulk tumors.

Methods

Problem formulation and notation

Let $X \in \mathbb{R}_{\geq 0}^{p \times n}$ denote the nonnegative gene expression matrix. DeSurv approximates $X \approx WH$, where $W \in \mathbb{R}_{\geq 0}^{p \times k}$ contains nonnegative gene programs and $H \in \mathbb{R}_{\geq 0}^{k \times n}$ contains sample loadings. Additionally, let $y \in \mathbb{R}_{>0}^{n}$ denote patient survival times, $\delta \in \{0,1\}^n$ the event indicators ($\delta_i = 1$ for an observed event, $\delta_i = 0$ for censoring), and $\beta \in \mathbb{R}^k$ the Cox regression coefficients. Survival outcomes are modeled through a Cox proportional hazards model with factor scores $Z = W^\top X \in \mathbb{R}^{k \times n}$: for sample i with projected score $z_i = W^\top x_i$, the hazard is $h(t \mid x_i) = h_0(t)\exp(z_i^\top \beta)$, where h_0 is the baseline hazard. Defining factor scores by projection onto the learned gene programs ($Z = W^\top X$) rather than by the inferred loadings H lets the same basis score new cohorts without refitting, the property we exploit for external validation.

The DeSurv Model

DeSurv integrates NMF with penalized Cox regression to identify gene programs associated with survival outcomes. The method operates in a survival-supervised framework: the reconstruction loss penalizes deviation from the observed expression data, while the survival term directs gene programs toward outcome-relevant structure. The parameter $\alpha \in [0, 1)$ controls the relative contribution of each objective.

The joint objective is

$$\mathcal{L}(W, H, \beta) = (1-\alpha)\,\mathcal{L}_{\mathrm{NMF}}(W, H) - \alpha\,\mathcal{L}_{\mathrm{Cox}}(W, \beta), \tag{1}$$

where $\mathcal{L}_{\mathrm{NMF}}(W, H)$ is the NMF reconstruction error (including an ℓ_2 penalty on H) and $\mathcal{L}_{\mathrm{Cox}}(W, \beta)$ is the elastic-net penalized partial log-likelihood. A key design choice is where survival supervision enters the factorization. Because factor scores are defined as $Z = W^\top X$, the Cox partial likelihood $\mathcal{L}_{\mathrm{Cox}}$ is a direct function of W and β, and the survival gradient acts explicitly on the gene program matrix W. The sample-level loadings H enter only through the reconstruction term and receive no survival gradient, preserving their interpretation as mixture coefficients[12,42]. When $\alpha = 0$, DeSurv reduces to standard unsupervised NMF, and β is fit post hoc by penalized Coxnet on the projected scores $Z = W^\top X$, used only for scoring and cross-validation without affecting the factorization. The Cox component also supports optional adjustment for sample-level covariates (for example, stage or grade); the analyses reported here use factor scores alone.

Optimization alternates updates for H, W, and β (Supplementary Information, Section 2). The reconstruction term is minimized using multiplicative updates that preserve nonnegativity; the supervision term enters W's gradient as a Cox partial-likelihood derivative; and β is updated by Coxnet coordinate descent on the current factor scores $Z = W^\top X$ with elastic-net penalty parameters (λ, ξ). Although the joint problem is non-convex, the idealized alternating scheme converges to a stationary point under mild regularity conditions, and the implementation includes practical modifications (Supplementary Theorem 1). Closed-form update equations, the convergence proof, and BO control parameters are given in Supplementary Information, Sections 1–3.

Hyperparameter selection and cross-validation

Hyperparameters $(k, \alpha, \lambda, \xi, n_{\text{top}})$ were selected by maximizing the cross-validated Harrell C-index using Bayesian optimization (BO)[43,44], which finds near-optimal configurations using far fewer objective evaluations than grid or random search over mixed integer–continuous hyperparameter spaces, with final rank k chosen by the one-standard-error rule (the smallest k whose predicted performance lies within one standard error of the maximum). All model selection was performed entirely within the training data using cross-validation; no validation cohort data were used during any stage of tuning, and unbiased generalization was assessed only on the external validation cohorts. Each fold was trained using multiple random initializations, and fold-level performance was defined as the average C-index across initializations. For stability, we used a consensus-based initialization for the final model, aggregating multiple DeSurv runs into a gene-gene co-occurrence matrix and constructing an initialization W_0 from the resulting clusters (Supplementary Information, Section 6). Before projection, each learned gene program (column of W) was restricted to its BO-selected number of top genes (denoted $\widetilde{W}$) and column-normalized (ℓ_2-norm scaling) by the same procedure used during training (Supplementary Information, Section 6); the resulting basis $\widetilde{W}$ was held fixed for all validation cohorts. In PDAC, BO selected $n_{\text{top}} = 270$ top genes per factor. The ridge penalties on the loadings (H) and gene programs (W) were held fixed at 0 in all analyses and were not tuned; factor scaling is instead controlled directly by column-normalization of W during optimization.

Because the goal is prognostic factorization, DeSurv selects k using the criterion that directly measures prognostic performance (cross-validated C-index) rather than the unsupervised diagnostics commonly used for NMF rank selection[13,25]. In the PDAC training data, standard NMF diagnostics (reconstruction residuals, cophenetic correlation, mean silhouette width) yielded inconsistent guidance across candidate ranks, a known limitation of unsupervised rank-selection criteria that can disagree on the same dataset (Supplementary Fig. S5). Cross-validated concordance varied across candidate ranks but plateaued from $k = 3$ through $k = 12$ (Supplementary Fig. S4). The one-standard-error rule applied to joint BO over rank and supervision strength accordingly selected the parsimonious $k = 3$ at $\alpha = 0.334$.

External validation cohorts were scored by projecting new datasets onto the learned programs via $Z = \widetilde{W}^\top X_{\text{new}}$, and survival associations were evaluated using Harrell's C-index, hazard ratios, and log-rank statistics; because Harrell's C-index is rank-based and assesses discrimination only, not calibration or time-dependent prediction error, we additionally report a proper score (the integrated Brier score and its index of prediction accuracy relative to a Kaplan–Meier reference) as a sensitivity check (Supplementary Information, Proper-scoring sensitivity). Because the survival-supervised information lives in W rather than H, scoring is a pure matrix projection requiring no retraining or validation survival information. Related supervised-NMF methods that instead route survival supervision through H require sample-loading re-estimation on each new cohort[28,29]. Validation survival outcomes were therefore used only for downstream performance evaluation, not for model training, scoring, or hyperparameter tuning. Reported external hazard ratios and log-rank statistics therefore reflect purely out-of-sample generalization. Further training and validation details appear in Supplementary Information, Part I.

Simulation studies

Simulation studies were conducted to assess recovery of prognostic latent structure and survival prediction under controlled conditions. Gene expression data were generated from a nonnegative factor model $X = WH$, where gene loadings W comprised three gene classes: marker genes, background genes, and noise genes. Marker genes were simulated to load strongly on a single factor and weakly on others, background genes to load strongly across all factors, and noise genes to have uniformly low loadings. Specifically, marker gene primary loadings were drawn from $\text{Gamma}(\text{shape} = 3, \text{rate} = 0.8)$ and cross-loadings onto other factors from $\text{Gamma}(1, 20)$; background gene loadings from $\text{Gamma}(2, 1)$ across all factors; and noise gene loadings from $\text{Gamma}(1, 20)$. Sample-level factor activities H were drawn from $\text{Gamma}(2, 2)$. Each dataset comprised $G = 3{,}000$ genes, $N = 200$ samples, and $K = 3$ factors, with 150 marker genes per factor and 500 shared background genes. Cox regression coefficients were $\beta = (2, 0, 0)^\top$ in the prognostic scenario, $\beta = 0$ in the null scenario, and $\beta = (2, 0, 0)^\top$ in the mixed scenario where survival risk was driven by 300 genes comprising the 150 factor-1 markers and 150 randomly sampled background genes, creating partial overlap between the prognostic signal and variance-dominant outcome-neutral programs.

Survival times were generated from an exponential distribution in which risk depended on marker gene expression through $X^T W_{\text{true}}$, where W_{true} retained marker gene loadings for their corresponding factors and was zero otherwise; censoring times were generated independently from an exponential distribution. Each dataset was analyzed using DeSurv and standard NMF followed by Cox regression on inferred factors, both tuned using cross-validated concordance index via Bayesian optimization. Performance was summarized across repeated simulation replicates. We considered three simulation scenarios: a prognostic scenario in which prognostic programs explain low variance relative to outcome-neutral signals, a null scenario with no survival signal, and a mixed scenario where prognostic and variance-dominant programs partially overlap.

Real-world datasets

We analyzed publicly available and controlled-access RNA sequencing (RNA-seq) and microarray cohorts of PDAC with corresponding overall survival outcomes. Eligible samples were primary PDAC tumors with a measured bulk expression profile and non-missing overall-survival time and event status; accrual and follow-up periods are as defined in each cohort's original publication (referenced in Supplementary Information, Section 7). Gene expression matrices were analyzed on a log scale: RNA-seq cohorts as $\log_2(\text{TPM} + 1)$; microarray cohorts in platform-normalized log-scale intensities. For each training cohort separately, genes were ranked by mean expression and by variance independently; the top 3,000 genes with the lowest sum of these two ranks were retained, selecting genes that are both highly expressed and highly variable; the intersection across training cohorts yielded 1,970 genes used for all analyses. Validation datasets were restricted to this same gene set. Survival times and censoring indicators were taken from the associated clinical annotations. No missing-data imputation was performed: all analyses used complete cases, and analyses adjusted for the PurIST and DeCAF classifiers were restricted to samples with available classifier calls. Of the seven PDAC cohorts we considered, two were used for training (TCGA and CPTAC) and five were used for external validation (Dijk, Moffitt, PACA-AU array, PACA-AU RNA-seq, and Puleo). To harmonize differences in scale across cohorts, filtered gene expression data were within-sample rank transformed before model training (ranks are inherently nonnegative, preserving NMF compatibility); ranking removes platform- and library-size differences in expression scale that would otherwise dominate cross-cohort projection. More details about the datasets can be found in Supplementary Information, Section 7.

Per-cohort participant flow is summarized in Supplementary Table S1. Of 321 pooled training-cohort samples (TCGA-PAAD, $n = 181$; CPTAC-3, $n = 140$), 48 were excluded for missing survival time, missing event indicator, missing tumor grade, or quality-control failure, yielding 273 analytic training samples (TCGA-PAAD 144, CPTAC-3 129; 139 events). Of 979 pooled validation-cohort samples, 363 were excluded for non-PDAC histology, non-tumor tissue, or missing survival annotation, yielding 616 analytic validation samples (414 events) across five cohorts: Dijk ($n = 90$, 81 events), Moffitt GEO array ($n = 123$, 83), PACA-AU array ($n = 63$, 38), PACA-AU RNA-seq ($n = 52$, 31), and Puleo array ($n = 288$, 181). These analytic samples provided ample events per estimated parameter: 139 training events for the 3-factor model (approximately 46 events per factor, well above the conventional minimum of ten) and 414 events across the external validation cohorts.

Data Availability

The training and validation datasets used in this study are publicly available or accessible through controlled-access repositories, and are de-identified; no Institutional Review Board (IRB) approval was required for their analysis. Training data were obtained from the Genomic Data Commons: TCGA-PAAD[45] and CPTAC-3[46]. Validation cohorts were obtained from ArrayExpress (Dijk, E-MTAB-6830[47]; Puleo, E-MTAB-6134[48]), the Gene Expression Omnibus (GEO; Moffitt, GSE71729[49]), and the International Cancer Genome Consortium (ICGC): the PACA-AU array cohort is publicly available from the Gene Expression Omnibus (GSE36924), and the PACA-AU RNA-seq cohort from the European Genome–phenome Archive (EGA) study EGAS00001000154 under controlled access[50]. Expression data and molecular classifications (PurIST, DeCAF) for the validation cohorts were originally compiled for the DeCAF study[5]; cohort details and sample sizes after quality control are summarized in Supplementary Information, Section 7. The treated bulk-tissue cohorts used in the exploratory translational analysis (Rash-Accept and the Linehan FOLFIRINOX$_{\pm}$CCR2-inhibitor series), and the COMPASS cohort used for the GATA6 RNA in situ hybridization comparison, are restricted-access clinical-trial datasets that are not publicly deposited; raw data are available from the respective study investigators under a data-use agreement, subject to the governing trial consents and institutional approvals. To preserve reproducibility without redistributing restricted data, the derived per-cohort summary statistics reported in the main text and Supplementary Information are provided in the reproducibility repository (file `results/treated_cohort_stats.rds`), so that every treated-cohort value can be inspected and verified.

Code Availability

All analyses were performed in R (version 4.6.0; R Core Team) using DeSurv (v1.0.1; this work), `NMF`[42], `glmnet`[51], `survival`, `ggplot2`, and `cowplot`; Bayesian optimization for hyperparameter selection was implemented within DeSurv. The DeSurv R package is available at https://github.com/rashidlab/DeSurv (archived at Zenodo, DOI `10.5281/zenodo.20150929`). Reproducibility code, processed data, and the exact session-info file are available at https://github.com/rashidlab/DeSurv-paper (archived at Zenodo, DOI `10.5281/zenodo.20150946`); this includes `code/11_treated_cohort_analysis.R`, which generates the treated-cohort summary statistics above.

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Inclusion and Diversity. The patient cohorts analyzed in this study were curated by the original investigators or consortia (TCGA, CPTAC, ICGC, and the original Moffitt, Dijk, and Puleo studies) and our access was limited to the de-identified expression and survival data they provide; we did not stratify analyses by sex, race, or ethnicity because consistent demographic annotations were not available across all cohorts. Sex and other demographic variables that could modify the prognostic relevance of the gene programs identified here remain important directions for follow-up work as more comprehensively annotated cohorts become available.

Author Contributions

Conceptualization: A.M.Y., N.U.R. Methodology: A.M.Y., A.Y., D.L., N.U.R. Software: A.M.Y. Formal analysis: A.M.Y. Investigation: A.M.Y. Data curation: A.M.Y., X.L.P. Visualization: A.M.Y. Writing – original draft: A.M.Y., N.U.R. Writing – review and editing: A.M.Y., A.Y., X.L.P., J.J.Y., D.L., N.U.R. Resources (PDAC domain expertise and biological interpretation): X.L.P., J.J.Y. Supervision: N.U.R. Funding acquisition: A.M.Y., N.U.R. All authors reviewed and approved the final manuscript.

Competing Interests

N.U.R. and J.J.Y. are named as inventors on US patents US20220127676A1 and US20250066856A1, which relate to pancreatic cancer subtype classification. The PurIST and DeCAF classifiers were used only as published, independently derived comparators: their subtype calls were obtained with the released classifiers and included as fixed covariate annotations, and the adjustment analyses using these calls are objective and reproducible from the deposited code. They are not components of DeSurv. The other authors declare no competing interests.

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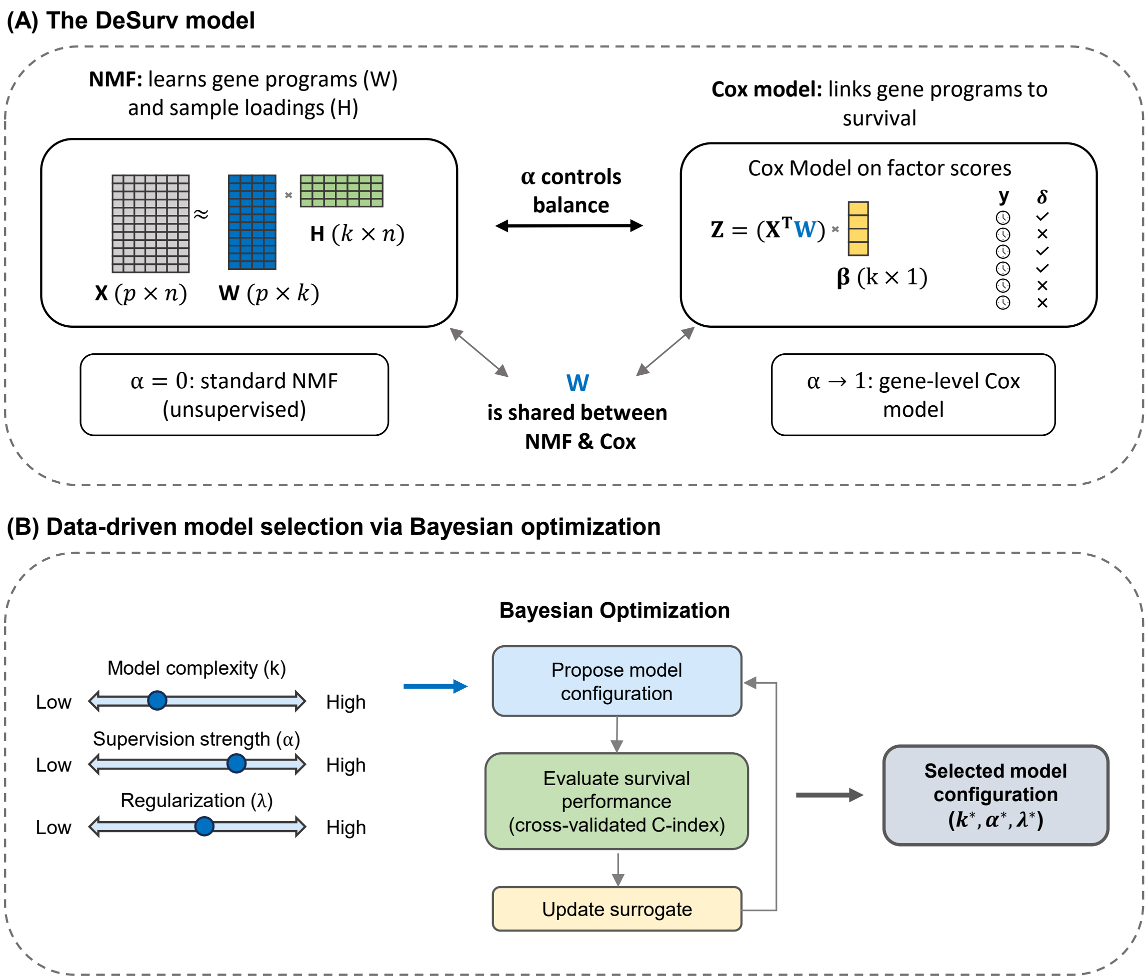

Figure 1: Overview of the DeSurv framework. (A) DeSurv jointly optimizes NMF reconstruction ($X \approx WH$) and Cox proportional hazards likelihood. The gene program matrix W is shared between objectives: factor scores $Z = W^\top X$ serve as covariates in the Cox model with coefficients β. A supervision parameter $\alpha \in [0, 1)$ controls the balance between reconstruction fidelity ($\alpha = 0$, standard NMF) and survival prediction (α close to 1). (B) The factorization rank (k), supervision strength (α), and regularization (λ) are selected via Bayesian optimization over cross-validated concordance index.

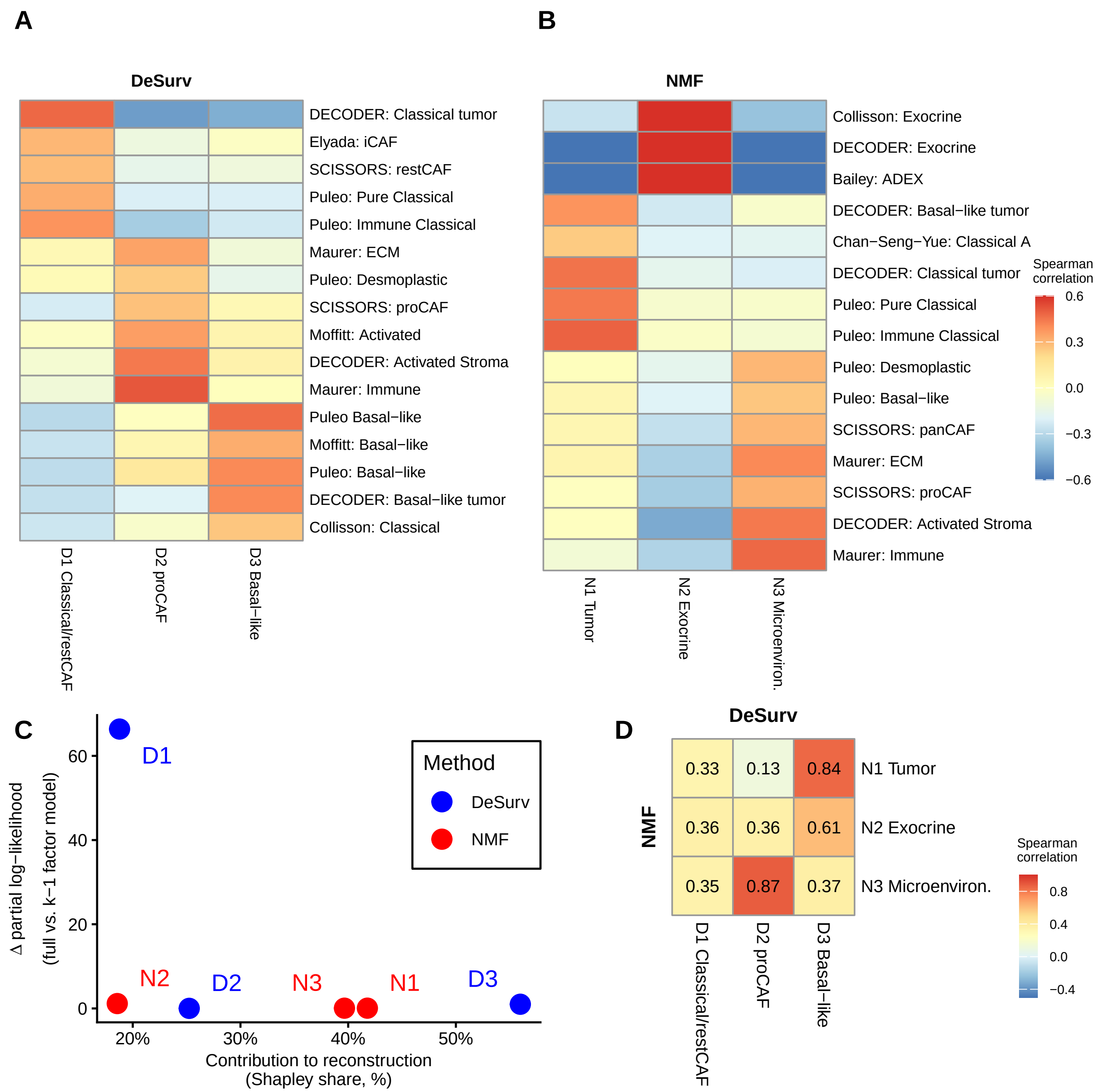

Figure 2: Survival supervision produces different factor structures in PDAC (TCGA + CPTAC training cohort, $n = 273$ patients, 139 events). (A,B) Spearman correlations between factor gene rankings (top 50 genes per factor) and established PDAC gene programs for DeSurv (A) and standard NMF (B) at $k = 3$; only $|r| > 0.2$ shown. (C) Per-factor reconstruction contribution (normalized Shapley share, summing to 100%) versus survival contribution (Δ Cox partial log-likelihood on factor removal). (D) Pairwise correspondence between NMF and DeSurv factors (Spearman correlation of W-matrix gene loadings over all genes).

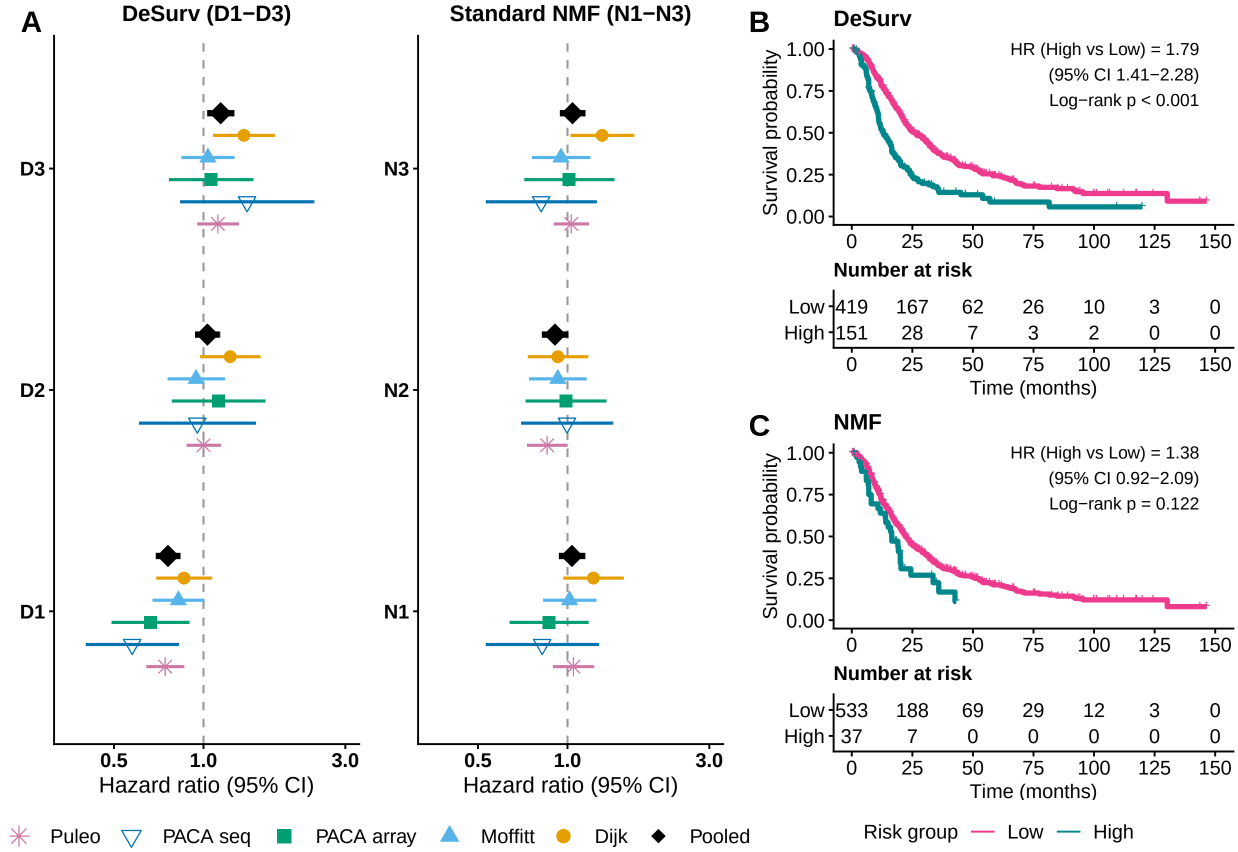

Figure 3: Survival-aligned programs generalize across independent PDAC cohorts (pooled $n = 616$ patients across 5 cohorts: Dijk $n = 90$; Moffitt GEO array $n = 123$; PACA-AU array $n = 63$; PACA-AU RNA-seq $n = 52$; Puleo array $n = 288$). (A) Forest plot of per-cohort univariate hazard ratios (95% CIs) for each factor; black diamonds denote inverse-variance-weighted pooled estimates. (B,C) Kaplan–Meier curves for pooled validation samples stratified by the training-derived multivariate linear predictor, dichotomized at a cross-validated optimal cutpoint: DeSurv (B) and standard NMF (C). No validation-cohort survival outcomes were used in training, scoring or hyperparameter selection.

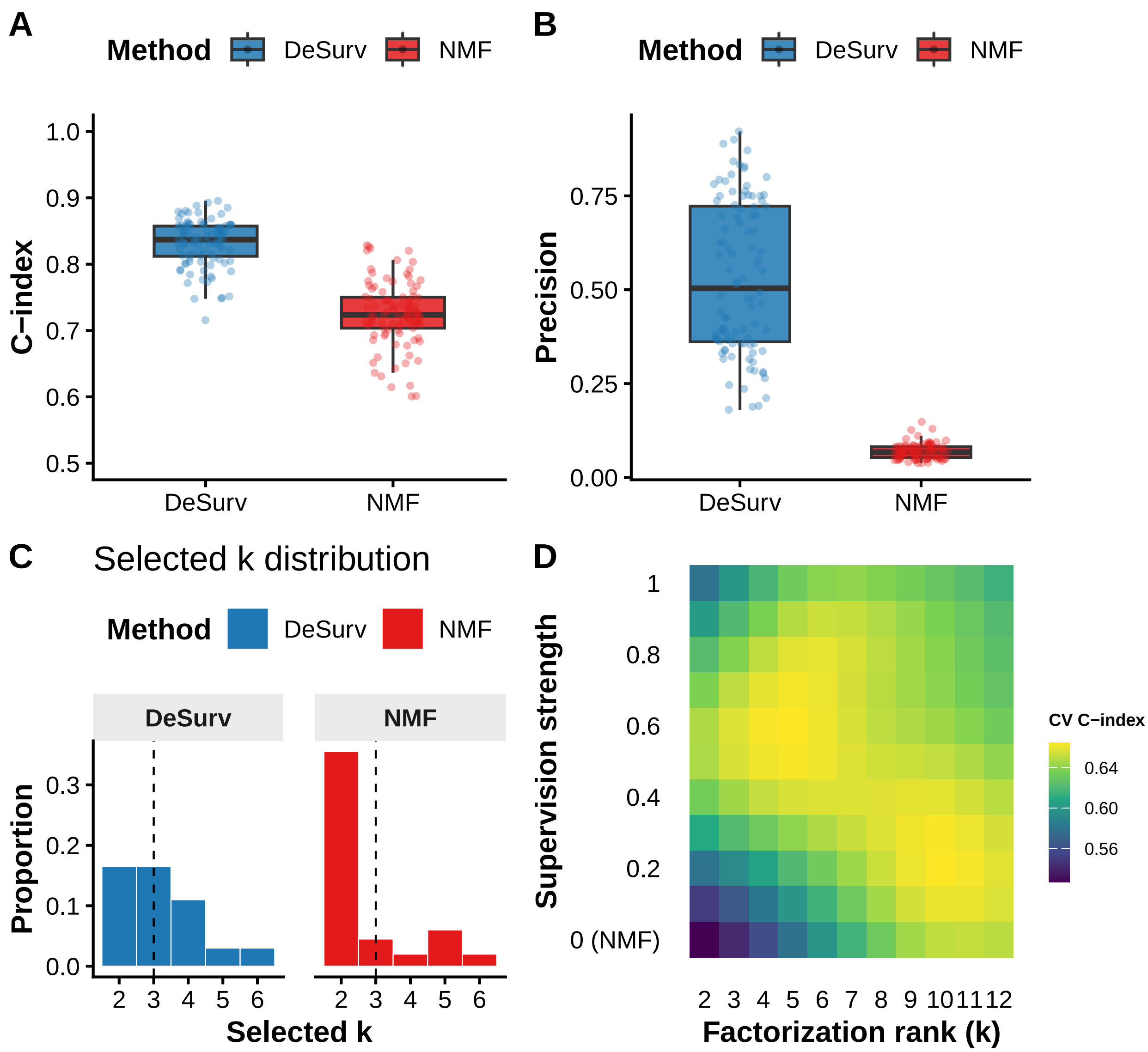

Figure 4: Survival supervision improves recovery of prognostic gene programs in simulations and PDAC training data. (A) Test-set concordance across 100 simulation replicates comparing DeSurv with standard NMF ($\alpha = 0$; $p = 3{,}000$ genes, $n = 200$ samples, true $k = 3$; paired Wilcoxon $P < 0.001$). (B) Precision of recovered prognostic gene programs (overlap between each true program and the top-weighted genes of its best-matching learned factor). In (A) and (B), boxes show median and interquartile range; whiskers extend to 1.5×IQR; points are individual replicates. (C) Distribution of selected ranks across replicates for DeSurv versus standard NMF. (D) Gaussian-process–predicted mean cross-validated C-index over factorization rank and supervision strength in PDAC training data (TCGA + CPTAC); this panel shows empirical tuning, not a simulation result.